

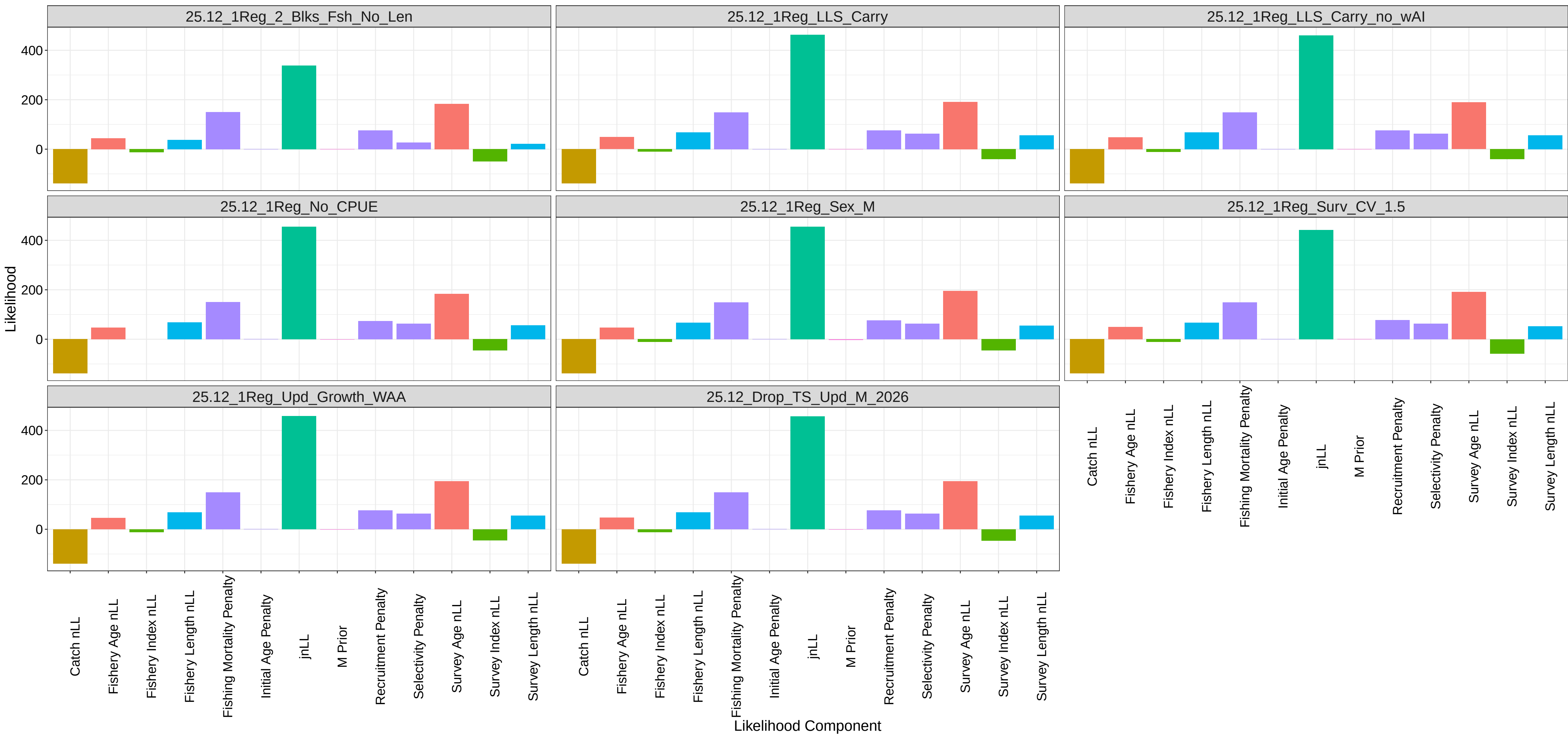
	Model	Region	Terminal_SSB	Terminal_SSB0	Terminal_F	Catch_Advice	B_Ref_Pt	F_Ref_Pt	B_over_B_Ref	B_over_DynB_Ref	F_over_F_Ref
1	25.12_Drop_TS_Upd_M_2026	1	206.6999	357.0208	0.04784668	39.55138	131.4549	0.07875	1.57240	0.57896	0.60758
2	25.12_1Reg_LLS_Carry	1	232.3958	357.0208	0.04244501	45.32573	135.5839	0.08029	1.71404	0.65093	0.52863
3	25.12_1Reg_LLS_Carry_no_wAI	1	214.4780	357.0208	0.04599081	41.56401	132.1464	0.07955	1.62303	0.60074	0.57813
4	25.12_1Reg_Surv_CV_1.5	1	224.2553	357.0208	0.04320235	44.22781	135.3843	0.07950	1.65644	0.62813	0.54342
5	25.12_1Reg_No_CPUE	1	221.0679	369.8268	0.04481018	41.69975	134.1211	0.07816	1.64827	0.59776	0.57329
6	25.12_1Reg_Sex_M	1	202.1604	364.8544	0.04964292	36.66894	137.0883	0.07555	1.47467	0.55409	0.65708
7	25.12_1Reg_Upd_Growth_WAA	1	204.2332	360.8372	0.04962997	37.82274	134.2543	0.07763	1.52124	0.56600	0.63928
8	25.12_1Reg_2_Blks_Fsh_No_Len	1	185.9384	340.5964	0.05264434	35.79575	127.2993	0.07813	1.46064	0.54592	0.67381

	Model	jnLL	M Prior	Recruitment Penalty	Selectivity Penalty	Catch nLL	Fishing Mortality Penalty	Survey Index nLL	Fishery Index nLL	Initial Age Penalty	Survey Age nLL	Fishery Age nLL	Survey Length nLL	Fishery Length nLL
1	25.12_Drop_TS_Upd_M_2026	456.8247	−1.0189779	75.86092	62.56423	−138.3008	149.3833	−45.39642	−11.21223	0.2741517	194.6365	46.66608	55.19863	68.16928
2	25.12_1Reg_LLS_Carry	463.4720	−0.6273865	76.03691	62.60458	−138.3019	149.3034	−40.12558	−10.14699	0.2519742	191.6554	48.93140	55.81667	68.07345
3	25.12_1Reg_LLS_Carry_no_wAI	459.7745	−0.7997904	75.69760	62.59432	−138.3219	149.2683	−40.17178	−10.69345	0.2622332	189.2432	48.21940	56.08357	68.39280
4	25.12_1Reg_Surv_CV_1.5	441.8561	−0.7751951	76.96210	62.64267	−138.1886	149.2995	−57.96658	−11.19367	0.2447329	190.9535	50.04496	52.78196	67.05065
5	25.12_1Reg_No_CPUE	455.7059	−0.9747952	73.15797	62.79468	−138.4294	150.4444	−45.63350	NA	0.1248466	183.4348	46.82344	55.77149	68.19207
6	25.12_1Reg_Sex_M	455.0521	−2.4573827	75.81780	62.57689	−138.2594	149.6971	−45.31139	−11.18961	0.2962465	195.1062	46.63170	54.70479	67.43926
7	25.12_1Reg_Upd_Growth_WAA	457.3437	−1.1262455	75.70519	62.61402	−138.2855	149.7030	−44.64577	−11.18826	0.2695543	193.8299	46.22126	55.51787	68.72868
8	25.12_1Reg_2_Blks_Fsh_No_Len	338.3738	−1.1576143	75.65710	27.19392	−138.4335	150.3366	−50.08440	−13.02572	0.3478061	183.3076	44.42418	22.22681	37.58098

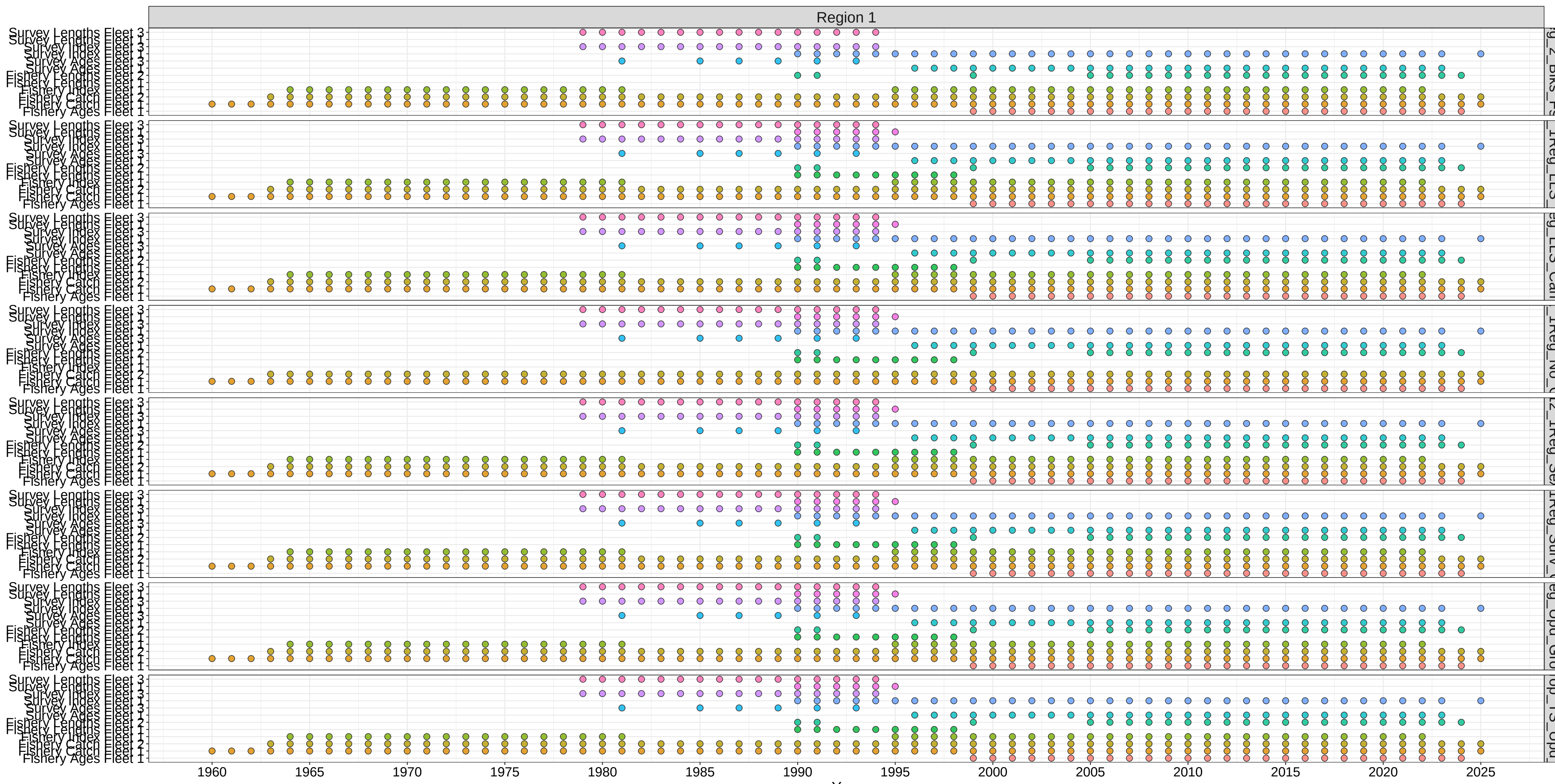
Negative Log-Likelihood

Type

Age	Index	Length	Prior
Catch	jnLL	Penalty	



Data



Natural Mortality

- Model
- 25.12_1Reg_2_Blks_Fsh_No_Len

25.12_1Reg_LLS_Carry_no_wAI

25.12_1Reg_Sex_M

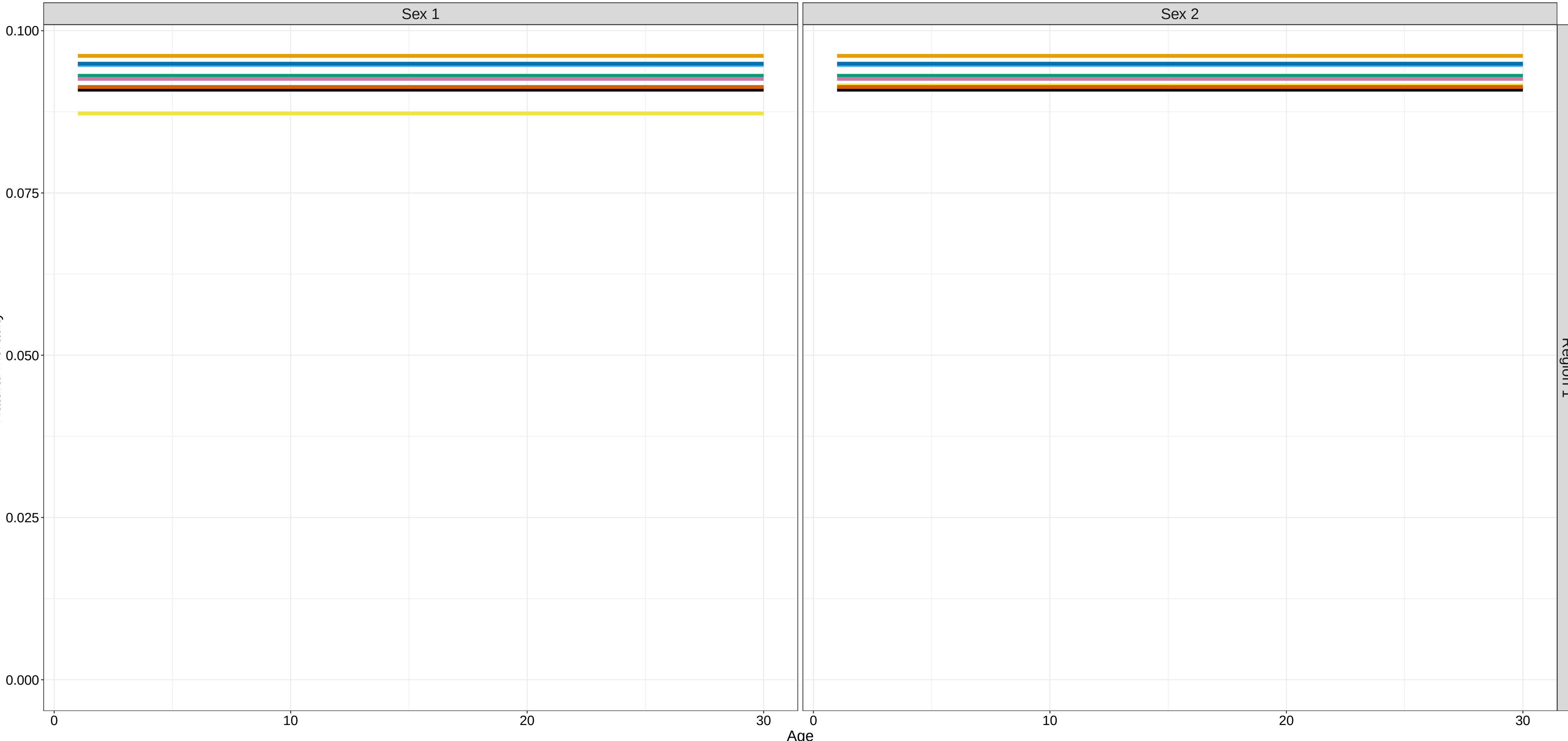
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25.12_1Reg_LLS_Carry

25.12_1Reg_No_CPUE

25.12_1Reg_Surv_CV_1.5

25.12_Drop_TS_Upd_M_2026



Weight-at-Age

- Model
- 25.12_1Reg_2_Blks_Fsh_No_Len

25.12_1Reg_LLS_Carry_no_wAI

25.12_1Reg_Sex_M

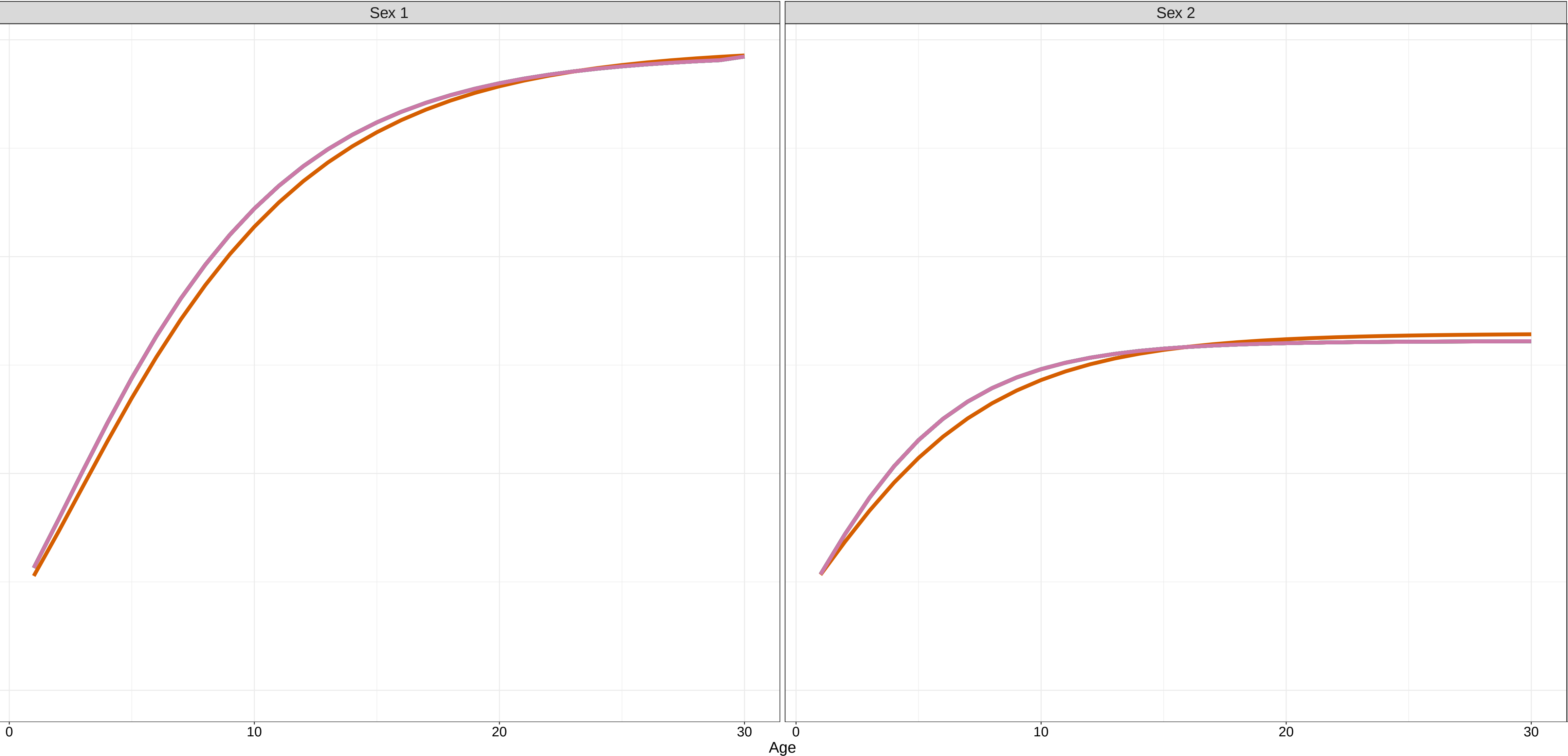
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25.12_1Reg_LLS_Carry

25.12_1Reg_No_CPUE

25.12_1Reg_Surv_CV_1.5

25.12_Drop_TS_Upd_M_2026



Maturity

- Model
- 25.12_1Reg_2_Blks_Fsh_No_Len

25.12_1Reg_LLS_Carry_no_wAI

25.12_1Reg_Sex_M

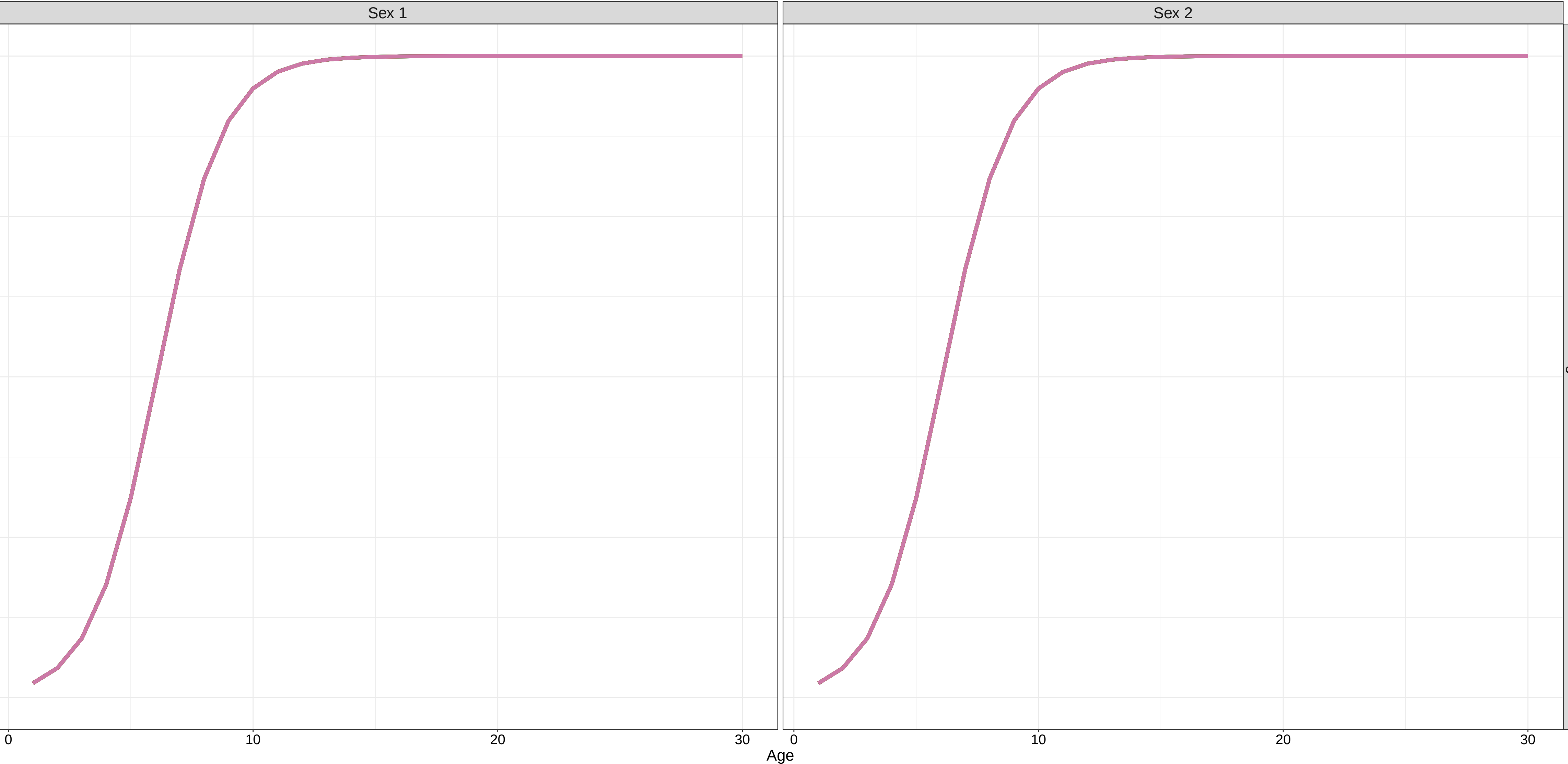
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25.12_1Reg_LLS_Carry

25.12_1Reg_No_CPUE

25.12_1Reg_Surv_CV_1.5

25.12_Drop_TS_Upd_M_2026



Index Fits

- Model
- 25.12_1Reg_2_Blks_Fsh_No_Len

25.12_1Reg_LLS_Carry

25.12_1Reg_LLS_Carry_no_wAI

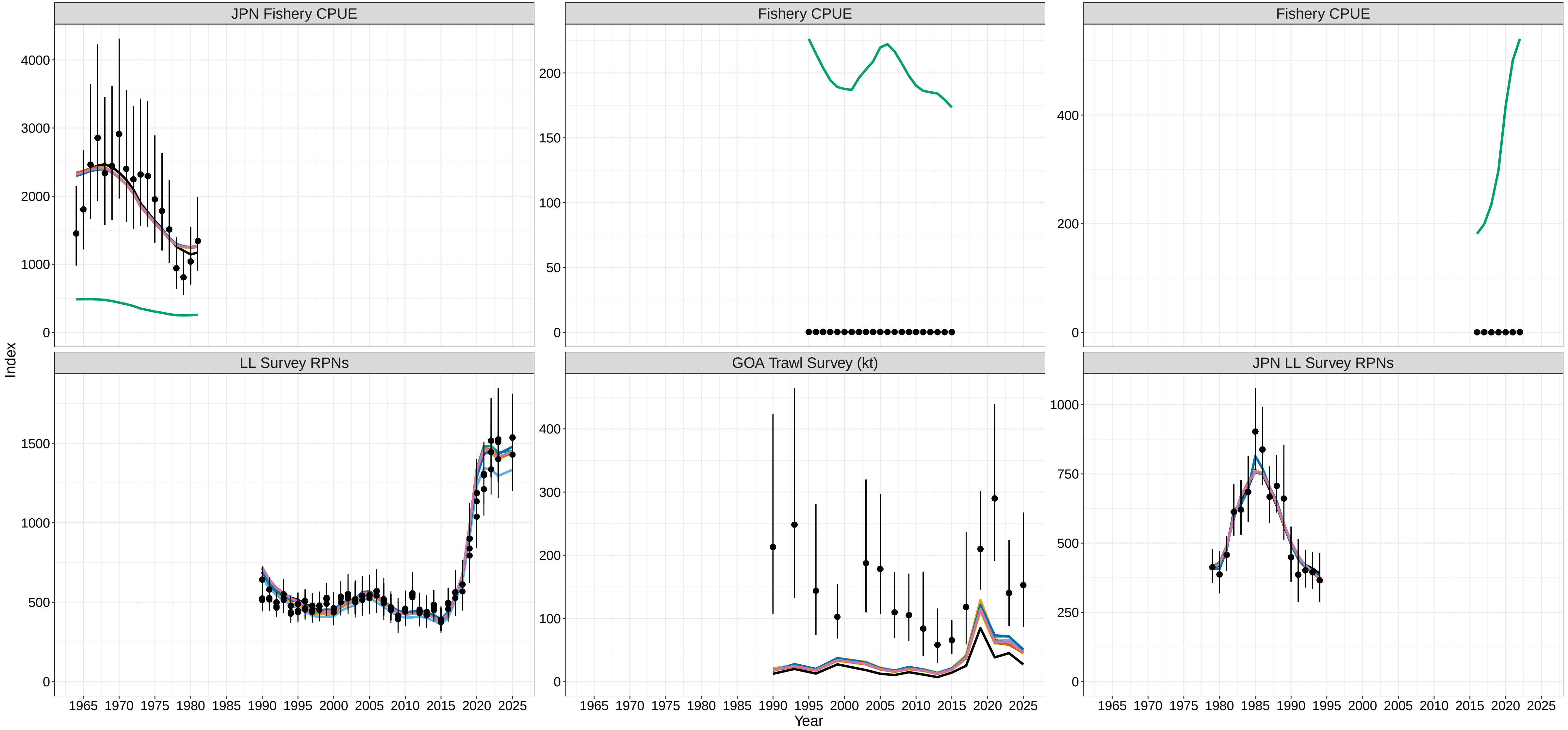
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25.12_1Reg_Sex_M

25.12_1Reg_Surv_CV_1.5

25.12_1Reg_Upd_Growth_WAA

25.12_Drop_TS_Upd_M_2026



Catch

- Model
- 25.12_1Reg_2_Blks_Fsh_No_Len

25.12_1Reg_LLS_Carry_no_wAI

25.12_1Reg_Sex_M

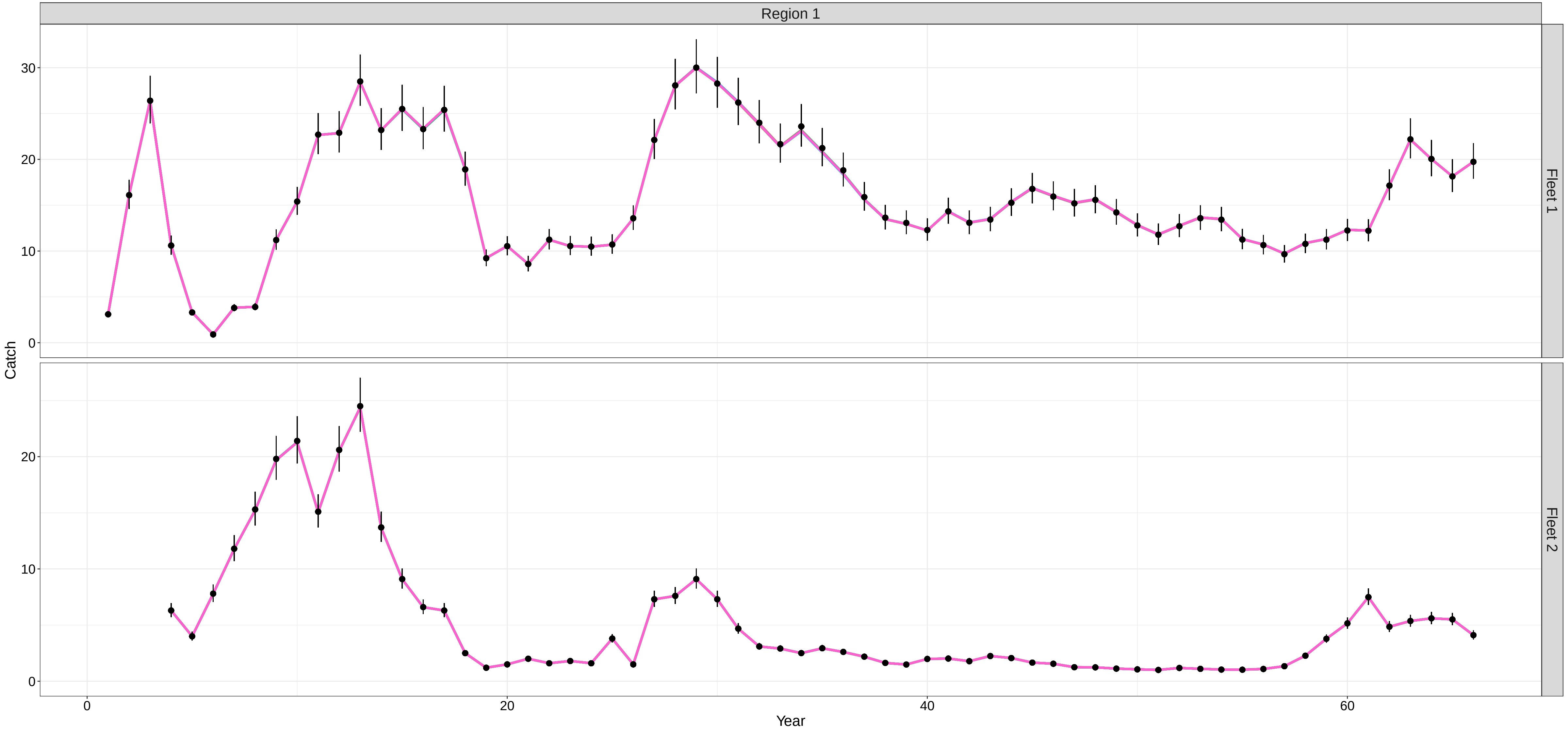
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25.12_1Reg_LLS_Carry

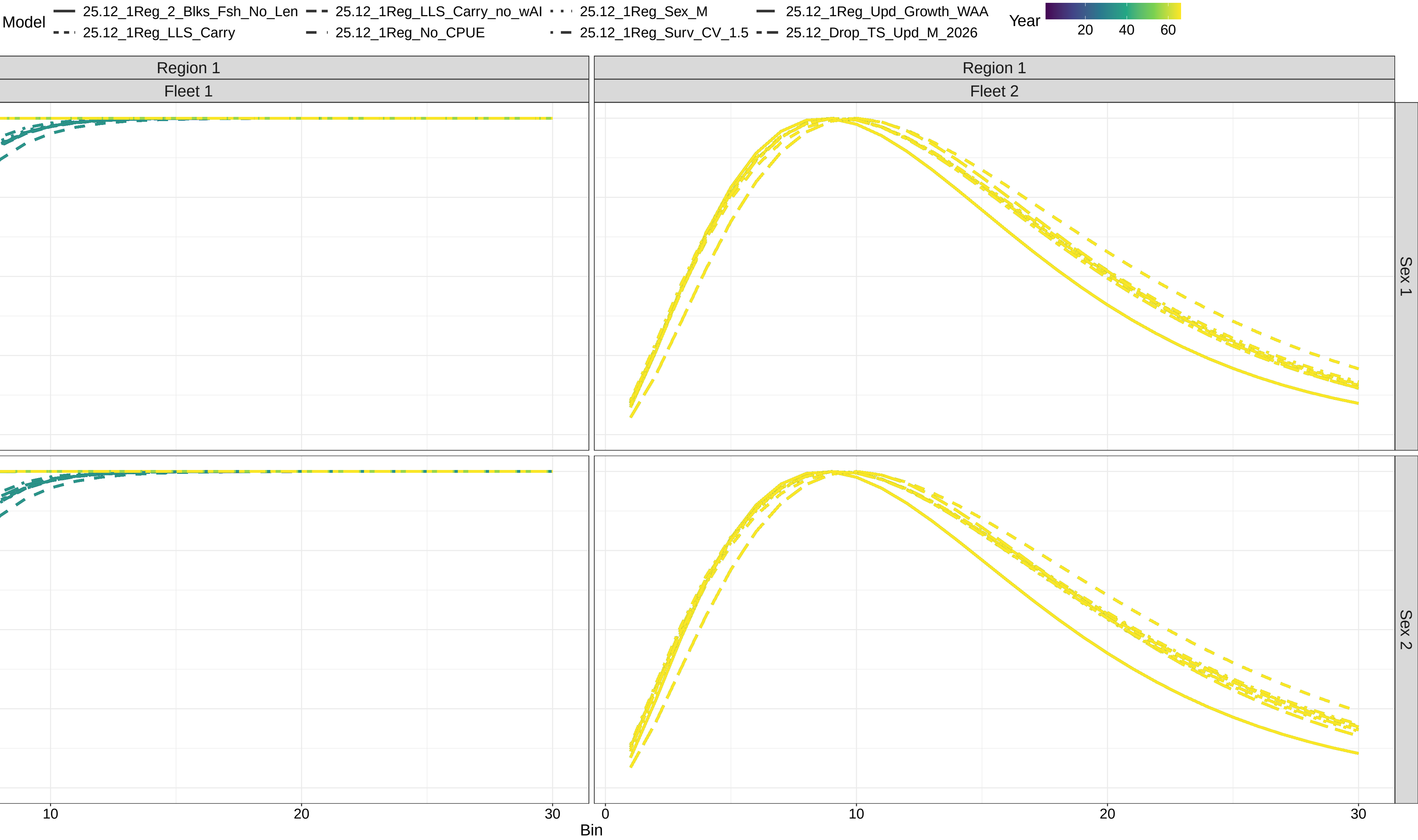
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25.12_1Reg_Surv_CV_1.5

25.12_Drop_TS_Upd_M_2026

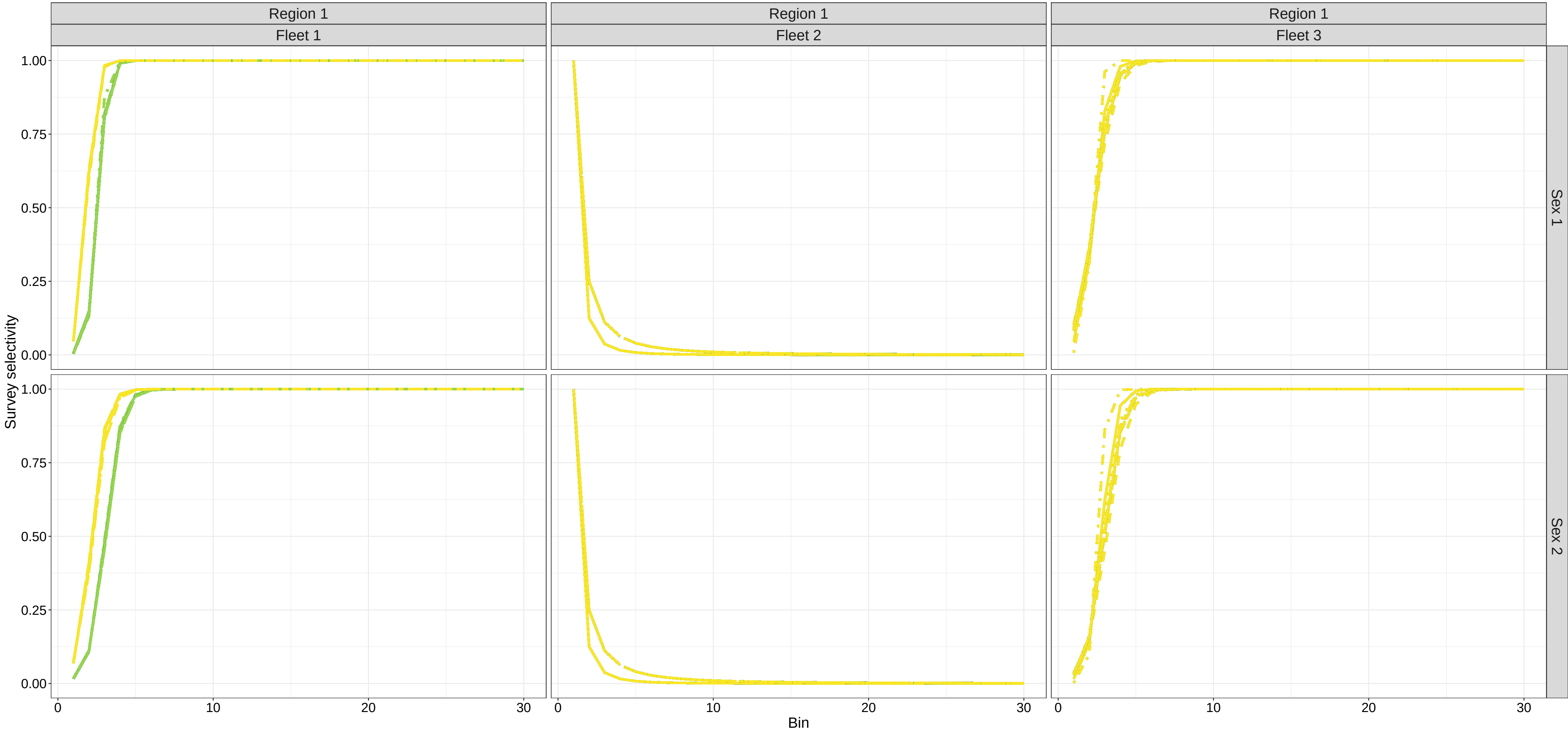


Fishery Selectivity



Survey Selectivity

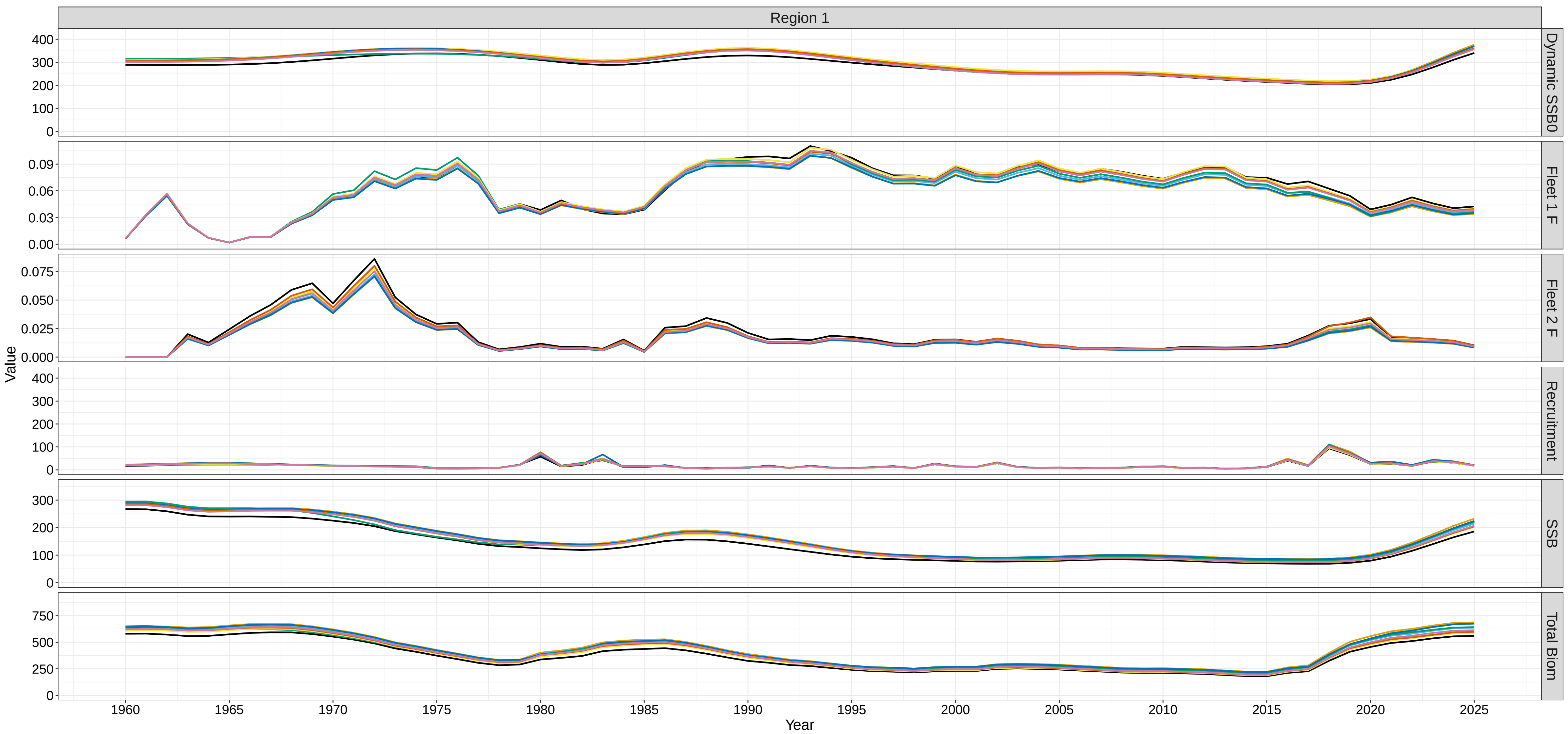
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- - · 25.12_1Reg_LLS_Carry - · 25.12_1Reg_No_CPUE · - 25.12_1Reg_Surv_CV_1.5 - - 25.12_Drop_TS_Upd_M_2026



Time Series Plots

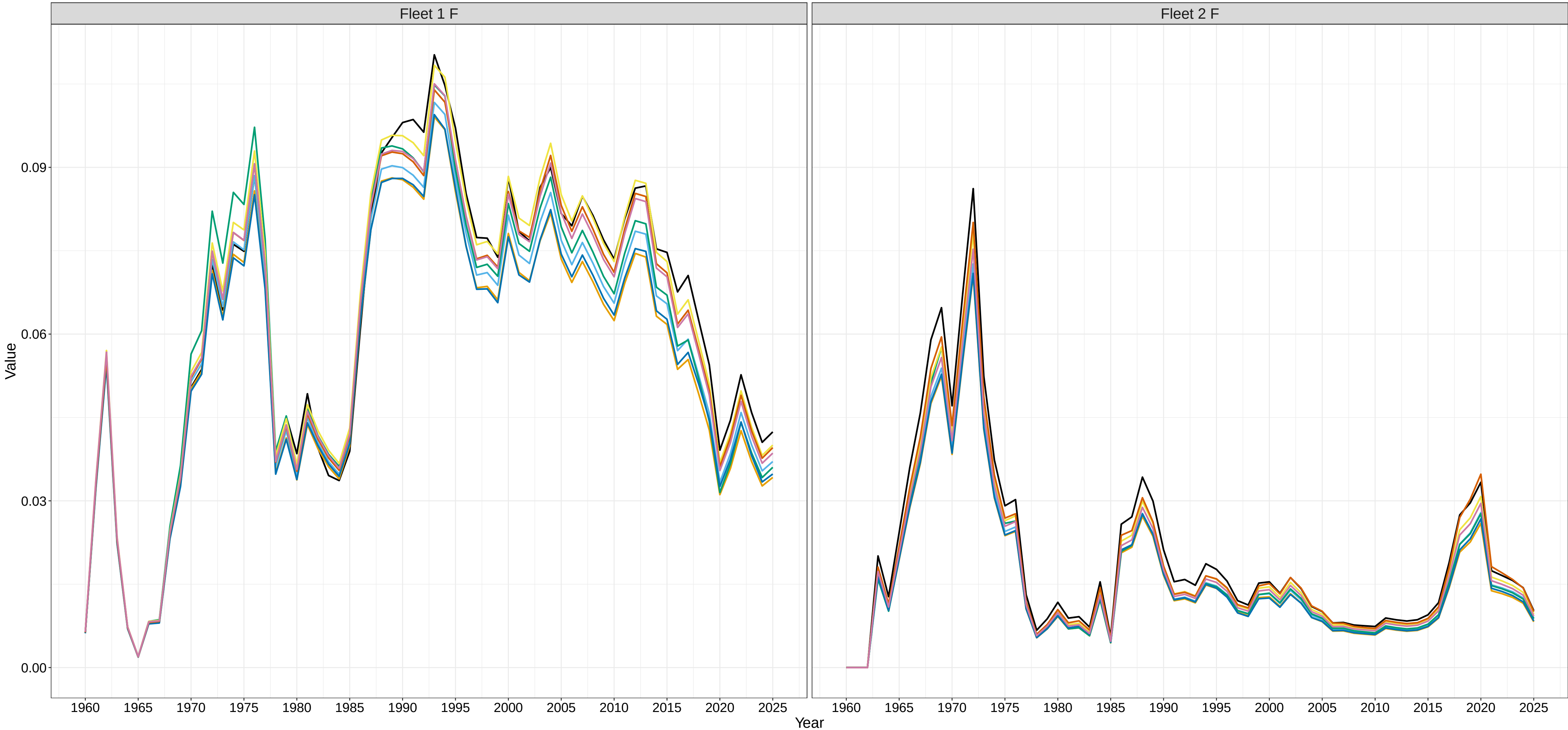
Model

- 25.12_1Reg_2_Blks_Fsh_No_Len
- 25.12_1Reg_LLS_Carry
- 25.12_1Reg_LLS_Carry_no_wAI
- 25.12_1Reg_No_CPUE
- 25.12_1Reg_Sex_M
- 25.12_1Reg_Surv_CV_1.5
- 25.12_1Reg_Upd_Growth_WAA
- 25.12_Drop_TS_Upd_M_2026



Fishing Mortality

- Model
- 25.12_1Reg_2_Blks_Fsh_No_Len
 - 25.12_1Reg_LLS_Carry_no_wAI
 - 25.12_1Reg_Sex_M
 - 25.12_1Reg_Upd_Growth_WAA
 - 25.12_1Reg_LLS_Carry
 - 25.12_1Reg_No_CPUE
 - 25.12_1Reg_Surv_CV_1.5
 - 25.12_Drop_TS_Upd_M_2026



Recruitment

Model

25.12_1Reg_2_Blks_Fsh_No_Len

25.12_1Reg_LLS_Carry_no_wAI

25.12_1Reg_Sex_M

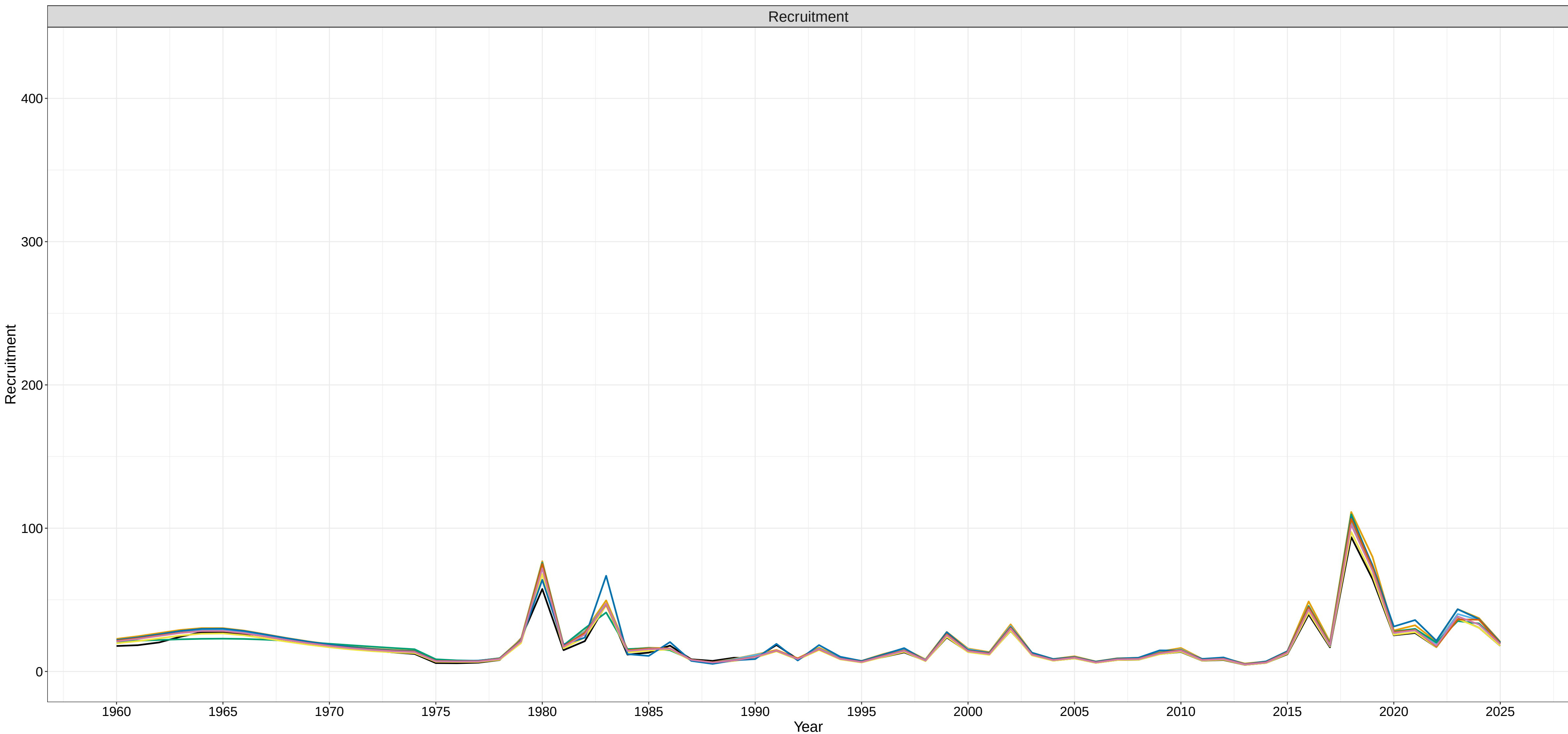
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25.12_1Reg_LLS_Carry

25.12_1Reg_No_CPUE

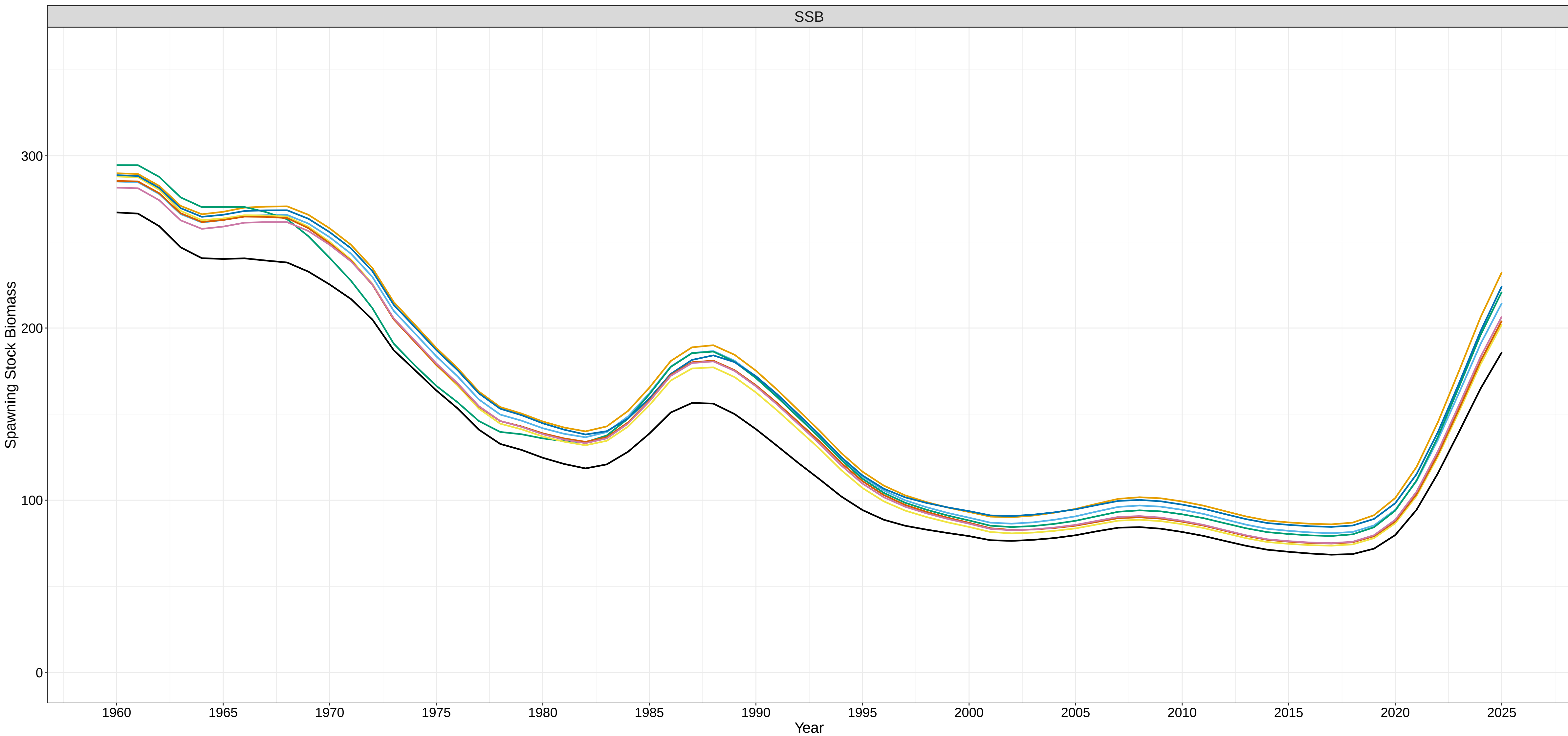
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25.12_Drop_TS_Upd_M_2026



SSB

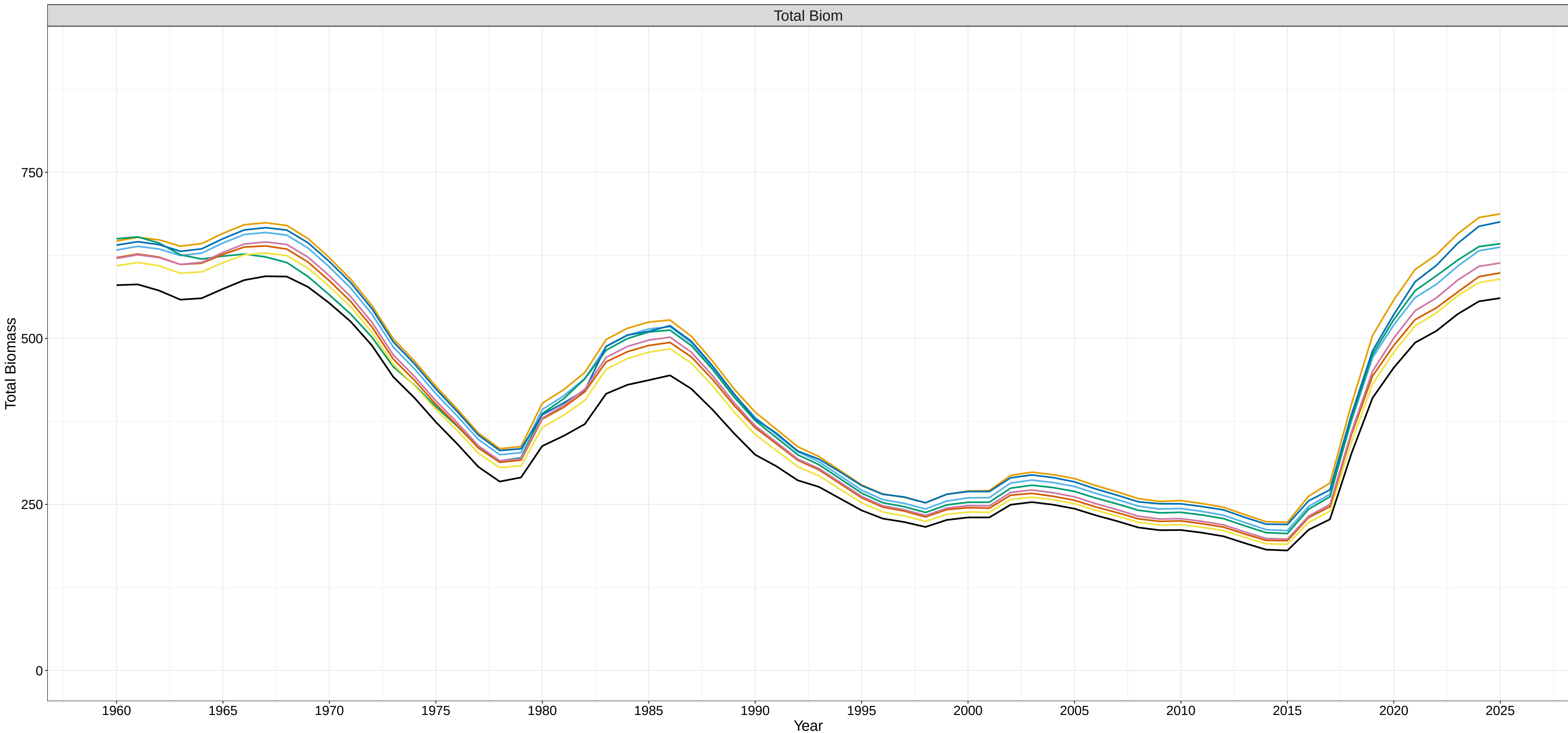
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— 25.12_1Reg_LLS_Carry — 25.12_1Reg_No_CPUE — 25.12_1Reg_Surv_CV_1.5 — 25.12_Drop_TS_Upd_M_2026



Biomass

Model

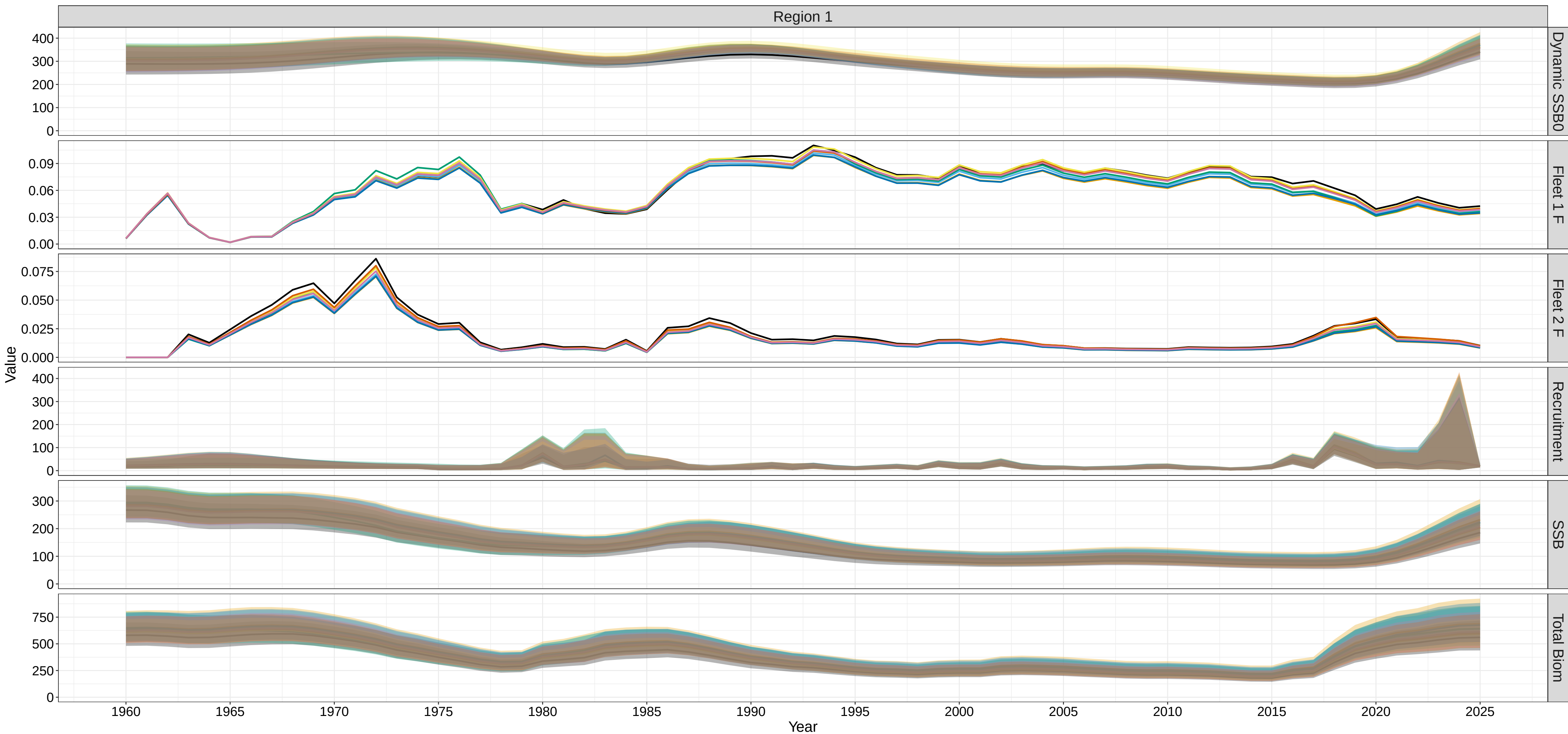
- 25.12_1Reg_2_Blks_Fsh_No_Len
- 25.12_1Reg_LLS_Carry
- 25.12_1Reg_LLS_Carry_no_wAI
- 25.12_1Reg_No_CPUE
- 25.12_1Reg_Sex_M
- 25.12_1Reg_Surv_CV_1.5
- 25.12_1Reg_Upd_Growth_WAA
- 25.12_Drop_TS_Upd_M_2026



Time Series Plots

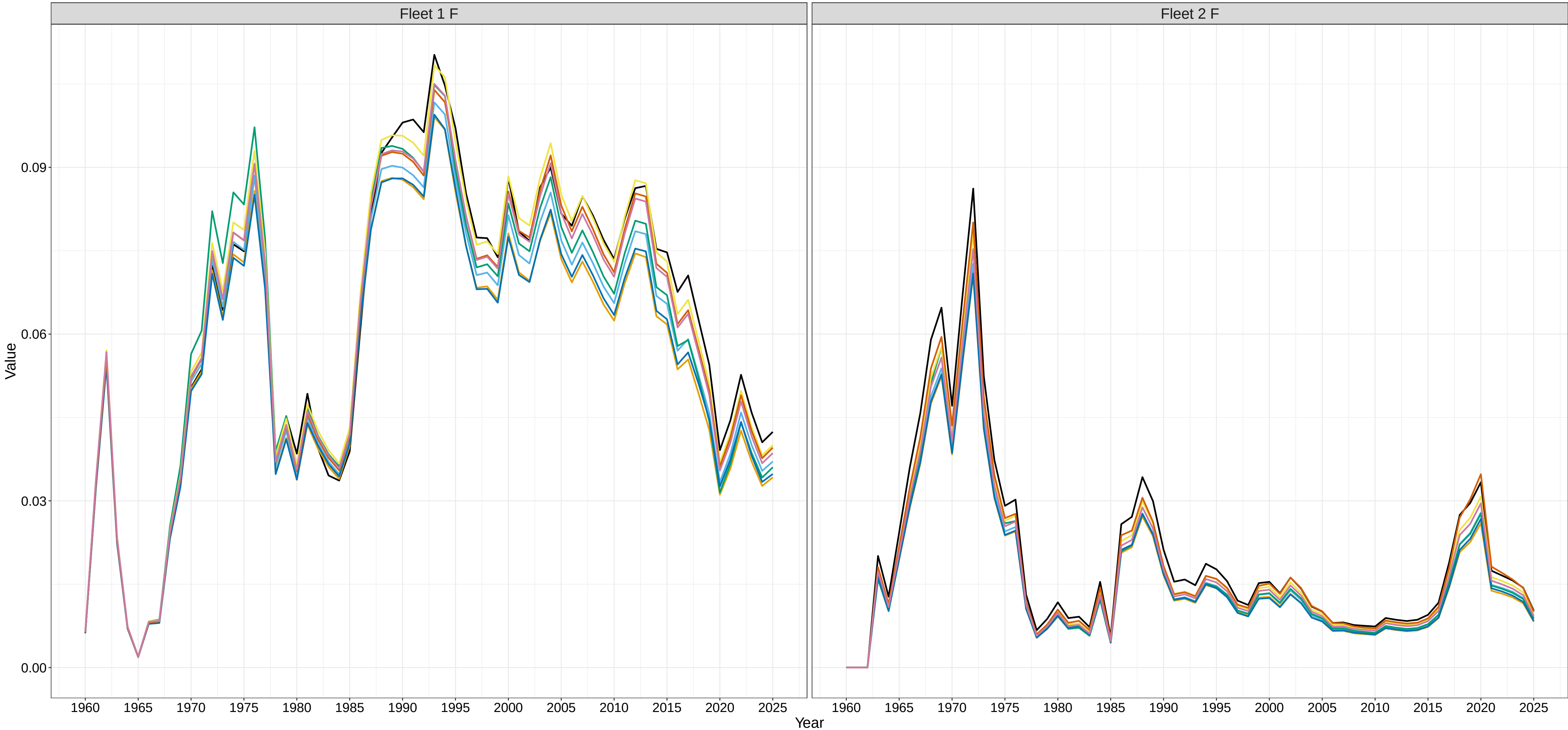
Model

25.12_1Reg_2_Blks_Fsh_No_Len	25.12_1Reg_LLS_Carry_no_wAI	25.12_1Reg_Sex_M	25.12_1Reg_Upd_Growth_WAA
25.12_1Reg_LLS_Carry	25.12_1Reg_No_CPUE	25.12_1Reg_Surv_CV_1.5	25.12_Drop_TS_Upd_M_2026



Fishing Mortality

- Model
- 25.12_1Reg_2_Blks_Fsh_No_Len
 - 25.12_1Reg_LLS_Carry_no_wAI
 - 25.12_1Reg_Sex_M
 - 25.12_1Reg_Upd_Growth_WAA
 - 25.12_1Reg_LLS_Carry
 - 25.12_1Reg_No_CPUE
 - 25.12_1Reg_Surv_CV_1.5
 - 25.12_Drop_TS_Upd_M_2026



Recruitment

Model

25.12_1Reg_2_Blks_Fsh_No_Len

25.12_1Reg_LLS_Carry_no_wAI

25.12_1Reg_Sex_M

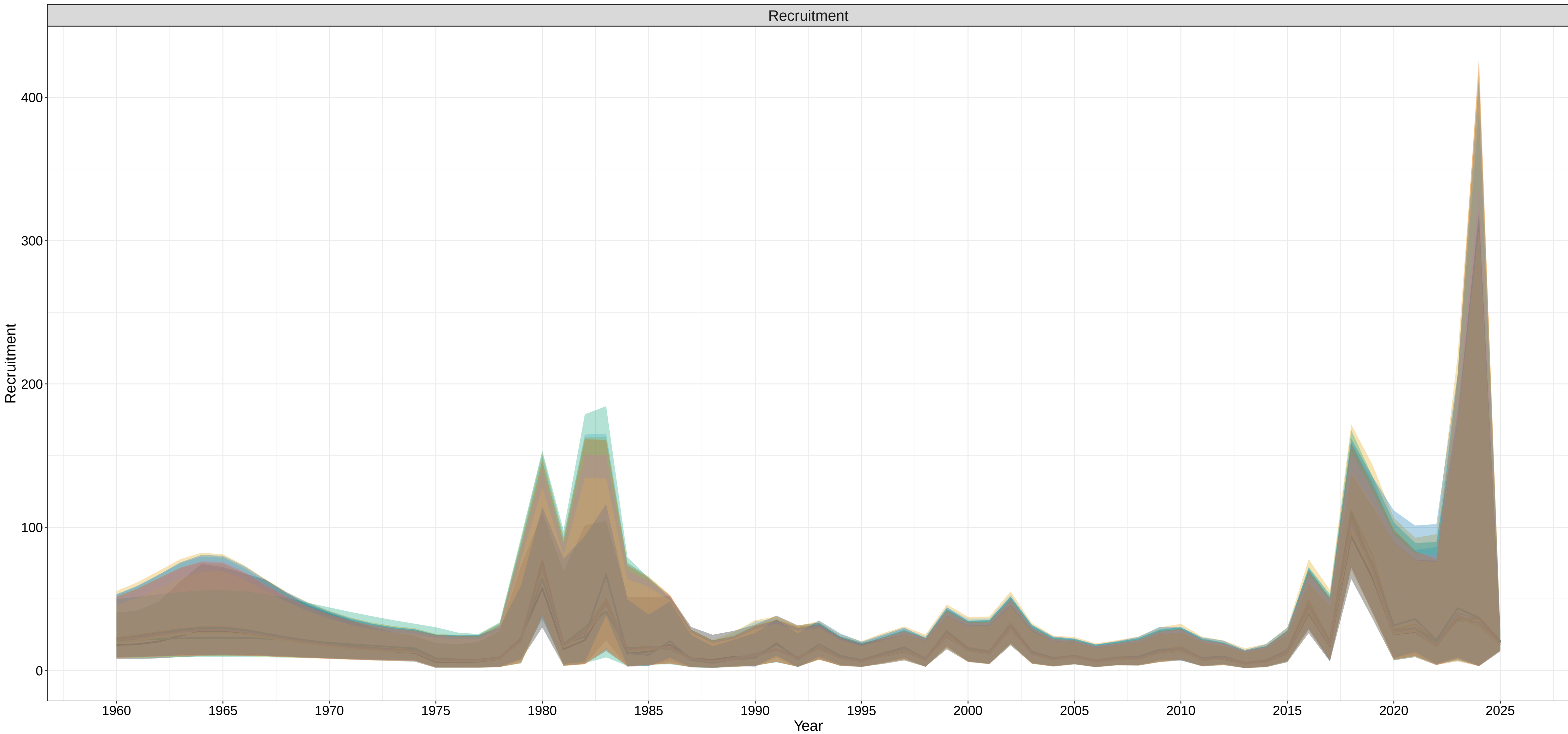
25.12_1Reg_Upd_Growth_WAA

25.12_1Reg_LLS_Carry

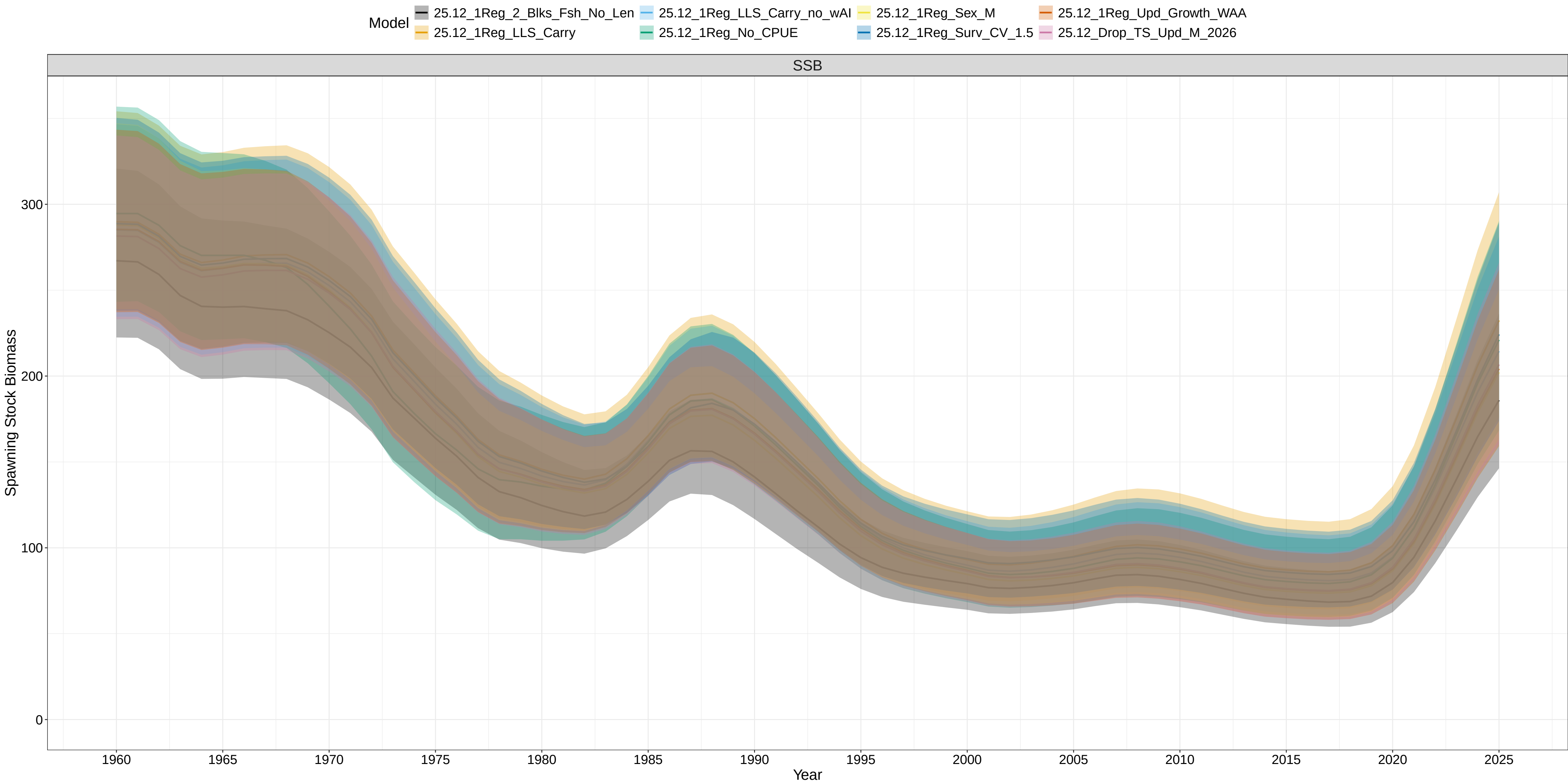
25.12_1Reg_No_CPUE

25.12_1Reg_Surv_CV_1.5

25.12_Drop_TS_Upd_M_2026



SSB



Biomass

Model

25.12_1Reg_2_Blks_Fsh_No_Len	25.12_1Reg_LLS_Carry_no_wAI	25.12_1Reg_Sex_M	25.12_1Reg_Upd_Growth_WAA
25.12_1Reg_LLS_Carry	25.12_1Reg_No_CPUE	25.12_1Reg_Surv_CV_1.5	25.12_Drop_TS_Upd_M_2026

